

1. Background & Research Questions

Sign languages are **inherently multi-channel (hands, body, and face)**, and some sign language models reflect this structure. We ask:

- How well do they **capture linguistic phenomena**?
- Do they use information across **multiple channels**?

2. ASL-MTP: ASL Minimal Translation Pairs

- 1,275 **ASL videos** with **matched and mismatched English translations** from ASLLRP
- 9 phenomena involving **a range of channel combinations**

Video Frames (ASL)



Transcription

He died city **o-s-a-k-a** (fingerspelled) on Tuesday <end>

Translation (English, **matched**)

He died in **Osaka** on Tuesday .

Minimal Pair Creation

He died in **Kyoto** on Tuesday .

Surprisal Analysis

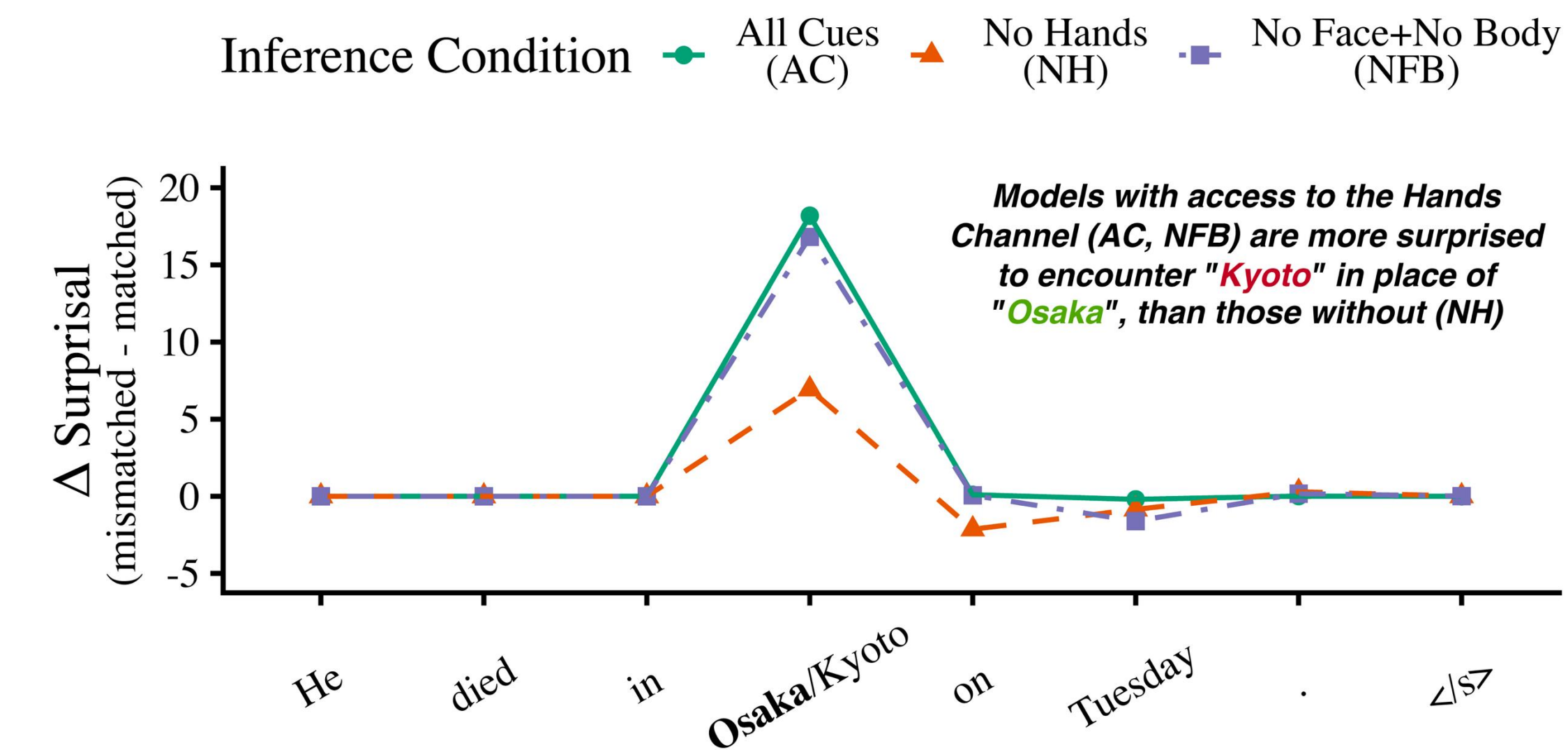


Figure 1. Our dataset construction and analysis approach.

4. Main Findings

Minimal pair analysis using ASL-MTP reveals that the SHuBERT-ByT5 translation model:

- Performs **above chance** across most phenomena.
- Is driven **primarily by manual cues (hands)**.
- Has **limited sensitivity to non-manual cues (facial and body movements)**.
- Exhibits a **declarative bias**.

Phenomenon	Primary Cue(s)	Secondary Cue	#	AC	NE	NM	NF	NFB	NH	NHM	NHF	NHB
1. Numbers	Hands		119	0.87	0.82	0.78	0.77	0.75	0.61	0.58	0.55	0.55
2. Fingerspelling	Hands		170	0.78	0.75	0.70	0.72	0.68	0.49	0.46	0.43	0.49
3. Classifiers	Hands		150	0.63	0.62	0.59	0.59	0.53	0.51	0.48	0.49	0.47
4. Wh-Questions	Hands + Brow lowered	Head shake	123	0.75	0.74	0.76	0.76	0.72	0.66	0.65	0.67	0.64
5. Negation vs. Positive	Hands + Head shake		104	0.80	0.77	0.77	0.77	0.71	0.69	0.69	0.68	0.69
6. Positive vs. Negation	Hands		104	0.65	0.68	0.62	0.66	0.53	0.34	0.34	0.28	0.29
7a. Conditionals	Hands + Brow raise	Head thrust, Body forward	155	0.70	0.72	0.66	0.70	0.72	0.56	0.51	0.54	0.59
7b. Conditionals (<i>NM only</i>)	Brow raise	Head thrust, Body forward	50	0.68	0.68	0.68	0.70	0.76	0.60	0.62	0.60	0.64
8. Declaratives vs. Polar Questions	Hands		150	0.97	0.97	0.97	0.96	0.96	0.96	0.95	0.95	0.95
9a. Polar Qs vs. Declaratives	Hands + Brow raise	Head forward	57	0.04	0.04	0.04	0.05	0.04	0.05	0.05	0.04	0.04
9b. Polar Qs vs. Declaratives (<i>NM only</i>)	Brow raise	Head forward	93	0.09	0.09	0.09	0.09	0.10	0.15	0.14	0.09	0.13

Table 1. Phenomenon-wise surprisal-derived accuracies across **inference conditions**. Accuracy values are **boldfaced** if they are significantly different from the accuracy on the 'All Cues' (AC) inference condition ($p < .05$).

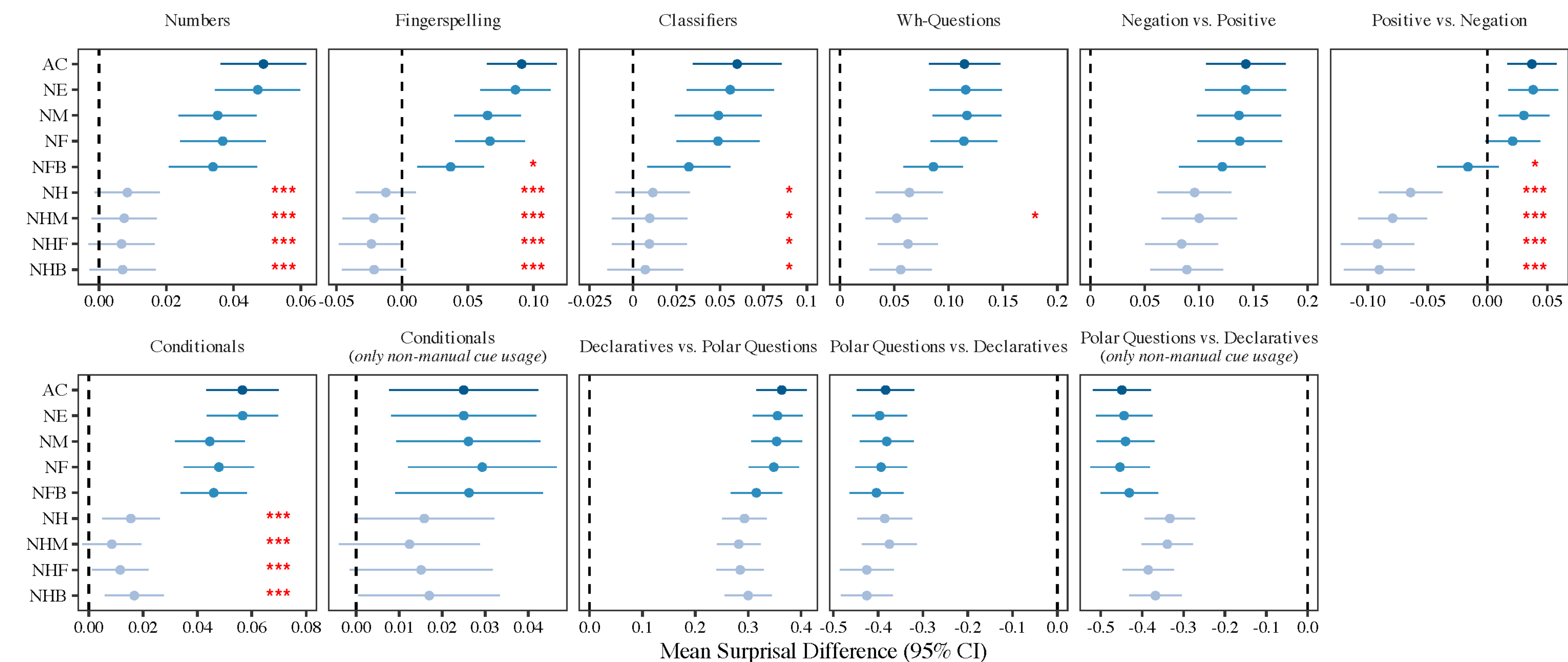


Figure 2. Average difference in surprisal of mismatched and matched sentences across phenomena and across **inference cue ablations**. *: $p < .05$; **: $p < .01$; ***: $p < .001$

3. Model

We test **SHuBERT-ByT5's** reliance on **multi-channel information** by manipulating channel availability at **training and test time**.

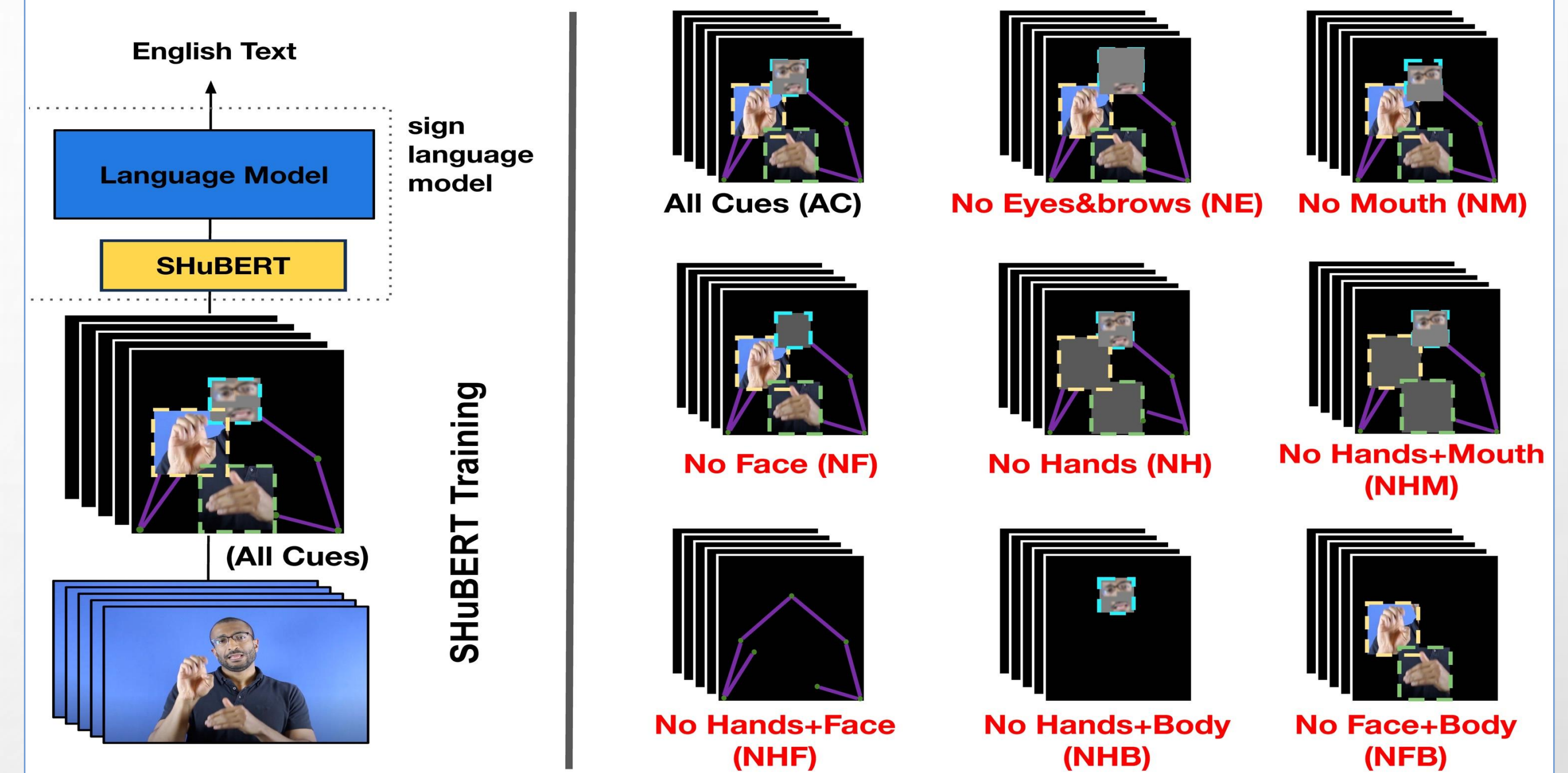


Figure 3. **Left**: SHuBERT is combined with an off-the-shelf language model (here, ByT5) to perform ASL-to-English translation. **Right**: Examples of inputs provided for the All Cues condition as well as the 8 Cue Ablations.

5. Additional Findings

BLEURT evaluation

- Minimal variation** across phenomena.
- Insensitive** to channel ablation effects.

Training ablations

- No consistent** performance improvements.
- Declarative bias** persists.

6. Conclusions

- ASL-MTP**: First minimal translation pair benchmark for sign language.
- Manual cue reliance**: Performance is driven largely by hand information.
- Inconsistent non-manual use**: Despite the multi-channel model, sensitivity to facial and body information is inconsistent.
- Fine-grained evaluation**: Minimal pairs uncover distinctions overlooked by aggregate translation metrics (e.g. BLEURT).